



Université
de Lomé

Domaine(s): Math, Physique, Chimie

Etablissement: Faculté Des Sciences (FDS)

Parcours: Licence

SYLLABUS DE COURS

Intitulé du parcours: Licence

Semestre d'évolution: Harmattan

Code et intitulé de l'enseignement: ANG200: Anglais Scientifique I

Nombre de crédits: 2

Enseignants responsables de l'UE: YOVO K. Koboè_ Docteur en Littérature de l'Afrique anglophone - ESP (Enseignant d'Anglais de spécialité)_ Université de Lomé: Département d'Anglais; ged.yovo@gmail.com

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Public cible: Etudiants

Prérequis: Avoir une notion en Anglais.

Objectifs de l'UE

Objectif général: Cette UE est une introduction à l'Anglais Scientifique, 1^{ère} Partie. Elle vise à aider l'étudiant à lire et à comprendre des articles publiés en Anglais en vue de pouvoir exploiter leurs contenus.

Objectifs spécifiques: A la fin de ce cours, l'étudiant doit être capable de:

- utiliser les mots ou expressions étudiés en vocabulaire dans leurs propres phrases.
- construire de bonnes phrases en anglais avec les structures syntaxiques étudiées.
- traduire correctement des passages des textes scientifiques étudiés en Français.
- répondre correctement aux questions basées sur les textes scientifiques étudiés.
- lire et écrire correctement les nombres entiers et non entiers, les heures et les dates, formules ou symboles mathématiques en Anglais.

Langues d'enseignement : Anglais et Français.

Bref descriptif de l'enseignement :

L'Anglais est l'une des langues les plus utilisées dans le domaine de la recherche. Ainsi, cette langue devient un outil linguistique indispensable qui s'impose pour tout acteur universitaire notamment les universitaires de la Faculté des Sciences. Cet enseignement vient donc satisfaire ce besoin linguistique.

Organisation de l'enseignement

Objectifs	Séance N°	Activités d'enseignement/apprentissage	Formules et techniques pédagogiques	Matériel/ Support pédagogique
Lancement de l'apprentissage	1	Présentation de la plateforme d'apprentissage à distance_Moodle - L'apprenant doit se connecter au site de l'UL et à la plateforme Moodle - Explorer l'environnement général de la plateforme Moodle Présentation du syllabus - Parcourir le syllabus - Echanger sur le syllabus	○ EXPOSE ○ SIMULATION ○ QUESTIONS-REPONSES	- Ordinateur ou Smartphone - Le support texte de cours - Kit de prise de notes
-construire de bonnes phrases avec les mots et expressions étudiés -utiliser les temps en Anglais -lire et écrire les nombres cardinaux en Anglais.	2	SESSION1: What is Science? -Construire une phrase avec chacun des mots soulignés dans le texte -Réviser les temps -Lire et écrire les nombres cardinaux.	GROUPE DE DISCUSSION ¹	- Ordinateur ou Smartphone - Le support texte de cours - Kit de prise de notes
-répondre aux questions -utiliser les temps -lire et écrire les	3	SESSION2: What is Science? (continued) - Répondre aux questions sur le	GROUPE DE DISCUSSION	- Ordinateur ou Smartphone

nombres cardinaux -Traduire un passage en Français/Anglais		texte - Réviser les temps (suite) - Lire et écrire les nombres cardinaux (suite) - Traduire le passage souligné en Français		- Le support texte de cours - Kit de prise de notes
-construire de bonnes phrases avec les mots et expressions étudiés -utiliser le conditionnel (If Clauses) - Lire et écrire les chiffres, les numéros de téléphone	4	SESSION3: The Mathematicians - Construire une phrase avec chacun des mots soulignés dans le texte -Étudier le conditionnel(If clauses) -Lire et écrire les chiffres, les numéros de téléphone	GROUPE DE DISCUSSION	- Ordinateur ou Smartphone - Le support texte de cours - Kit de prise de notes
-répondre aux questions -utiliser since-for-ago -lire et écrire les nombres ordinaux -Traduire un passage en Français/Anglais	5	SESSION4: The Mathematicians (continued) - Répondre aux questions sur le texte - Étudier since-for-ago - Lire et écrire les nombres ordinaux - Traduire le passage souligné en Français	GROUPE DE DISCUSSION	- Ordinateur ou Smartphone - Le support texte de cours - Kit de prise de notes
-construire de bonnes phrases avec les mots et expressions étudiés -utiliser les voix passives et actives -lire et écrire les fractions	6	SESSION5: What is Chemistry? - Construire une phrase avec chacun des mots soulignés dans le texte -Étudier les voix actives et passives -Lire et écrire les fractions	GROUPE DE DISCUSSION	- Ordinateur ou Smartphone - Le support texte de cours - Kit de prise de notes
- vérifier les enseignements reçus en ligne -s'évaluer	7	Regroupement présentiel de mi- semestre: - Echanger sur les activités effectuées en ligne, - Remédiation, - Evaluation de mi-parcours (DST) Prendre les consignes pour la suite des enseignements	○ GROUPE DE DISCUSSION ○ QUESTIONS-REPONSES	- Cahiers d'examen - Stylo - Epreuves
-répondre aux questions -utiliser le comparatif -lire et écrire les nombres décimaux	8	SESSION6: What is Chemistry? (continued) - Répondre aux questions sur le	GROUPE DE DISCUSSION	- Cahiers d'examen

¹ Échange de propos entre les apprenants sur un sujet donné et pour un temps déterminé avec ou sans la participation ou la modération de l'enseignant.

-Traduire un passage en Français/Anglais		texte -Étudier le comparatif - Lire et écrire les nombres décimaux - Traduire le passage souligné en Français		- Stylo - Epreuves
- construire de bonnes phrases avec les mots et expressions étudiés -utiliser le superlatif -dire les expressions désignant: nombre de fois, les pourcentages	9	SESSION7: The Different Areas of Physics - Construire une phrase avec chacun des mots soulignés dans le texte -Étudier le superlatif - Lire les expressions désignant: nombre de fois, les pourcentages	GROUPE DE DISCUSSION	- Ordinateur ou Smartphone - Le support texte de cours - Kit de prise de notes
-répondre aux questions -utiliser le gérondif -demander, dire et écrire l'heure -Traduire un passage en Français/Anglais	10	SESSION8: The Different Areas of Physics (continued) - Répondre aux questions sur le texte - Étudier le gerondif - Demander, dire et écrire l'heure - Traduire le passage souligné en Français	GROUPE DE DISCUSSION	- Ordinateur ou Smartphone - Le support texte de cours - Kit de prise de notes
- Traduire un passage en Français/Anglais -répondre aux questions -exprimer un contraste -demander, dire et écrire la date	11	SESSION9: SOLAR POWER -Traduire le passage souligné en Français -Répondre aux questions sur le texte - Étudier although/though; despite/in spite of -Demander, dire et écrire la date	GROUPE DE DISCUSSION	- Ordinateur ou Smartphone - Le support texte de cours - Kit de prise de notes
-vérifier les enseignements reçus en ligne -consolider les enseignements reçus -s'évaluer - faire l'examen final	12	Regroupement présentiel de fin de semestre: - Echanger sur les activités effectuées en ligne, - Remédiation, - Evaluation finale (Examen)	o GROUPE DE DISCUSSION o QUESTION S-REPOSES	- Cahiers d'examen - Stylo - Epreuves

Évaluation

- **Évaluation en cours d'apprentissage:** DST : 40%

- **Examen final:** Examen écrit: 60%

Bibliographie

- 1-Défourneaux, Marc and Défourneaux Roldan, Michelle. (1980) *Do You Speak Science?* Bordas: Paris.
- 2-Rodney, Huddleston and Geoffrey, K. Pullum. (2002) *The Cambridge Grammar of English Language*. Cambridge: Cambridge University Press.
- 3- EPOU K. A. and FAGNON K., *My Key to Success in English*. (unpublished)
- 4-<https://learnenglish.britishcouncil.org/english-grammar-reference/comparative-and-superlative-adjectives> retrieved on 23-11-2020.
- 5-http://guidetogrammar.org/grammar/tenses/future_perfect_progressive.htm retrieved on 24-11-2020.
- 6-https://www.englishclub.com/grammar/verb-tenses_future-continuous.htm retrieved on 24-11-2020.

Session1-Study of scientific text: WHAT IS SCIENCE?

VOCABULARY: Explain the underlined words through definition, synonym, antonym or translation.

eg. **Broad** = general, large; **link** = connect, relate, form a partnership

scientific method==>méthode scientifique; **hard sciences** ≠ **soft science**; **proponent** ≠ **detractors**

Practice

Build a meaningful sentence with each of the following words: broad (adj), link(v), proponent (n)

TEXT:

Science is a **broad** grouping of disciplines containing many different areas that are all **linked** together by a single concept-the scientific method. **The scientific method** represents an investigative method based on observation, deduction, hypothesizing, and experimentation that can be applied to all areas of life. But what are some of the individual types of science that **rely on** this method? Though there are many ways to look at science, one of the most common is to divide it into three broad categories, each of which contains numerous **subdisciplines**: formal science, natural science, and social science. Each of these types of science will be explained in more detail below.

Formal science represents those disciplines that deal with symbols and theoretical ideas and their applications in the real world. Its **inclusion** as a Science is often contested, but aspects of

formal science are used in all other scientific disciplines. Formal science includes computer science, mathematics, and statistics. Natural science is the science that people usually think of when they hear the term. Those studying it use the scientific method to understand nature and the physical world.

Natural science and its subdisciplines are sometimes referred to as “**hard Sciences**” by their **proponents**. Natural Science includes biology, chemistry, geology, and physics. Social science is the study of societies and the interactions within them, be they on a group or individual basis. Social science is sometimes referred to as a «**soft Science**» by **detractors**. Social science includes anthropology, psychology, and sociology. Each broad scientific category contains many disciplines and subdisciplines with specific research foci.

Grammar: REVIEW OF ENGLISH TENSES

A-Present Tenses

a-Simple present

- It expresses habitual and frequent actions.

Subject + inf. without “to” (3rd pers.+verb+s; 3rd pers.+verb-o/s/h/x+es; 3rd pers.+verb-y →cons.+ies/vow.+y+s)

eg.: I wake up at 6’ o’clock everyday.

- It expresses actions or events that happen at the time now.

eg. The student **copies** the English lesson.

- It expresses general truths.

eg. Babies **cry** when they are hungry.

- It expresses Eternal truth

The sky **is** above. /The earth **is** round. /The sun **rises** in the east.

- Proverbs or sayings are expressed in simple present tense.

eg. Patience **is** a golden road. /Walls **have** ears.

NB: Adverbs that are often with the present simple: always, generally, usually, occasionally, sometimes.

eg. She **always prays** before going out.

b-The present continuous

It describes an action taking place at the time of speaking

Subject + present of to be + verb + ing

eg. Students **are taking** notes in their note books

c-The present perfect

It describes an action which began/ started in the past ant that continues in the present time. (Emphasis is laid on the results)

Subject + has /have + past participle of V

eg.It has rained. /They have slept.

d-The present perfect continuous

It describes an action that recently stopped or is still going on; a finished action that influenced the present putting emphasis on the course or duration (not the result).

Subject + have or has + been + verb + ing

eg. She has been thinking. /She has been cooking.

B-Past Tenses

e-The simple past

Subject + verb (in past)

The simple past is used to say that something happened, took place and was definitely finished or completed in the past.

eg. I cried. / Anita went to that market yesterday.

Sub + preterit of Verb

Notes:

- The simple past is used to tell stories:
 - Once upon a time, there lived a hunter in the forest ...
 - One day, a man who farmed by the riverside...
 - The simple past is used with expressions like “**It is high time**”/ “**It is about time**”.
- Used for saying that something should be done soon, because it is already past the time when it should have been done.

eg. **It is high time** you **did** something.

f-Past continuous

The past continuous tense is used to indicate action that was still happening at a specific moment in the past when another event occurred or happened. It also describes simultaneous past actions in progress.

Subject + past of to be + verb + ing

e.g.: 1-They **were sleeping** while their sister **was watching** TV.

2-Michael **was trying** to take notes when his pen fell on the ground.

TRACTICE

Put the verbs in brackets into appropriate tenses

- a. The boy (cry) because he is hungry
- b. The sun always (to shine) in Mango
- c. Koffi is interested in politics but he (not/belong) to a political party
- d. Look at the river. It (to flow) very fast today
- e. Have you (to drive) a car before?
- f. Afi generally (to sing) in French but today she (to sing) in English
- g. She (be) a pilot before?
- h. Ali has never (to see) a crocodile
- i. Yesterday, I (wake) up at seven and (have) my breakfast at half past
- j. Last weekend, Jomo (to play) while we (to do) our homework.
- k. He (to work) in this bank since 2000

Mathematical tools: CARDINAL NUMBERS

This section requires from students to say numbers and mathematical expressions orally, and to write them correctly in letters/words or in numbers in English.

CARDINAL NUMBERS (Part1)

1-12 13-19 30-90 100- 200

1 one one (a) hundred

2 two two hundred

3 three 13 thirteen 30 thirty

4 four 14 fourteen 40 forty

5 five 15 fifteen 50 fifty

6 six 16 sixteen 60 sixty

7 seven 17 seventeen 70 seventy

8 eight 18 eighteen 80 eighty

9 nine 19 nineteen 90 ninety

10 ten 20 twenty

11 eleven 21 twenty-one

12 twelve 22 twenty-two

Be careful with the spellings: ***thirteen, fifteen, eighty, forty, and fifty***

345 = three hundred and forty-five.

Students should learn to construct sentences using cardinal numbers.

Sample sentences

1) The criminal spent a/one hundred days in prison.

2) I told the boy a hundred times to be careful.

3) There are 700,000 people without jobs.

4) My cell phone costs 50,000 francs.

PRACTICE

Write the following in figures

- 1) two million three hundred and seventy
- 2) five thousand six hundred and eighty-eight
- 3) four hundred and forty-five
- 4) fifty- five thousand three hundred and thirty

Session2-Reading Comprehension (WHAT IS SCIENCE?)

Read the text carefully and answer the questions.

QUESTIONS

- 1) In science we have several disciplines that can be grouped in three categories. Name them.
- 2°) Provide the subdisciplines that we have in each of the above categories.
- 3°) What do all these disciplines have in common?
- 4) The common feature of all these disciplines can be summarized in four words. Name them.

Grammar: REVIEW OF ENGLISH TENSES (continued)

g-Past perfect

This tense describes an action taking place before a certain time in the past.

Subject + had+ past participle of verb

eg.: She **had won** the competition.

It is used to show two actions that took place in the past and one action was finished first. We put the action which finished first into the past perfect tense and any other action in the same sentence into the simple past tense

eg. -The couple **had taken** their dinner by the time their friend **came**.

-When the Headmaster **arrived**, the meeting **had** already **started**.

h-Past perfect continuous

This tense is used to indicate an action that was continuing at a specific time in the past.

Subject + had been + verb + ing

eg.: The twin sisters **had been living** in USA for a year before they decided to move to France. /He **had been waiting** for you.

C-Future Tenses

i-Simple future

We use the simple future tense to talk about actions or events that will happen in the future time.

Subject + will/ shall + inf - to

eg.: They **will win** that match. /We **shall travel** to Paris for the next holidays.

j-The future continuous tense

This tense is a verb tense that indicates that something will occur in the future and continue for an expected length of time.

Subject + will/shall + be + verb + ing

eg. -**we'll be having** dinner when the film starts.

-He **will be playing** tennis at 10am tomorrow./ What **will** you **be doing** at 10pm tomorrow?

k-The future perfect

This tense is used to convey an action that will have been completed by a certain time in the future.

Subject + shall/will+ have + past participle of the verb

eg. -I **shall have written** six pages by to night. /-The sun **will have set** by seven o'clock.

-We have not yet met the English teacher but we **shall have done** so by evening.

l-Future perfect progressive

It indicates a continuous action that will be completed at some point in the future.

Subject + have + been + verb + ing

eg. -*When* I finish this course, I will have been learning English for nine years.

-By 2022 they **will have been living** in London for sixteen years.

-*Next year* he **will have been working** here for four years.

-The students **will have been doing** the exercise before the teacher enters the class.

PRACTICE

Mathematical tools: CARDINAL NUMBERS (continued)

Expressing a large indefinite number, eg we can say hundreds, thousands or millions.

Example sentences

- 1) Hundreds of football fans watched the match last week.
- 2) There are thousands of students at the University of Lome.

Other Whole Numbers

1000= a/one thousand, 3000= three thousand (singular).

1100 = eleven hundred etc.

In the following numbers note that there is a comma after every three digits 15,961 = fifteen thousand nine hundred and sixty-one.

842,372 = eight hundred forty-two thousand three hundred and seventy-two.

1,000,000 =one/a million, 2,000,000 =two million (singular).

333,555, 413 = three hundred and thirty-three million five hundred and fifty-five thousand four hundred and thirteen.

Approximate numbers are expressed using words such as ‘around, about, approximately’; eg in about ten days, around 11 weeks, Tsevie is **approximately 30 km** from Lome.

Extreme Numbers

0 = Nought, Zero

Infinity = ∞

TRANSLATION: Translate the passage in italic into French

Session3-Study of scientific text: THE MATHEMATICIANS

VOCABULARY: Explain the underlined words through definition, synonym, antonym or translation.

eg. Setting ==> time and space; allowed = permit; applied = practical; employment $\neq$ unemployment; deductive logic ==> logique déductive; intuitive leaps ==> démarches intuitives

Practice:

Build good sentences with each of the following: apply(v), setting(n), deductive logic

TEXT:

*Mathematicians are people who specialize in the study of mathematics, the manipulation of numbers, variables, and abstract concepts in equations and theorems. They can work on a variety of topics, from the history of math to theoretical mathematics, and they work in many industries, including computer science and engineering. Mathematicians are also an important part of the research teams which work on issues like geology, demographics, physics, and so forth, and they can be found in a variety of **settings**.*

(...) Mathematicians were responsible for calculating the data which **allowed** people to understand how the seasons worked, how to build huge structures, and how to perform a variety of other tasks. They also contributed immensely to the advancement of the sciences in many cultures.

Some mathematicians specialize in **applied** or **practical** mathematics, which involves the application of mathematics to real world issues. A statistician who compiles **employment** statistics, for example, could be considered an applied mathematician. (...)

Theoretical mathematicians work on the theory, rather than the practical application, of math. For example, a mathematician can use math to think of an entirely new way to approach

computer science, or to describe the dynamics of crowds. Theoretical mathematicians are usually found working at the university level.

Most mathematicians have a doctoral degree, and many have completed post-doctoral studies as well. They can work in academic settings as well as in offices of government agencies and other companies, and they may utilize a variety of tools in their work, including **computer systems** which are capable of very complex and detailed calculations. **Deductive logic** and **critical thinking skills** are also very important for professionals in this field, as is the ability to make **intuitive leaps**, in the case of people who work with theoretical math. (...)

Grammar: IF - CLAUSES TYPE I, II AND III

A. Conditional Sentence Type 1:

→ *It is possible and also very likely that the condition will be fulfilled.*

Form: *If* + Simple Present ==> Future (= will + bare infinitive)

Eg: **If I find** her address, **I'll send** her an invitation.

B. Conditional Sentence Type 2:

→ *It is possible but very unlikely, that the condition will be fulfilled.*

Form: *if* + Simple Past => Conditional (= would + bare infinitive)

Eg: **If I found** her address, **I would send** her an invitation.

C. Conditional Sentence Type 3 :

→ *It is impossible that the condition will be fulfilled because it refers to the past.*

Form: *if* + Past Perfect ==> Conditional II (= would + have + Past Participle)

Eg: **If I had found** her address, **I would have sent** her an invitation.

PRACTICE

Complete the Conditional Sentences by putting the verbs into the correct form.

1. Peggy **will go** shopping if she **(have)** time in the afternoon.
2. If they **(study / not)**harder, they **wouldn't pass** the exam.
3. If you **had asked** me, I **(help)**you.
4. If she **took the** red bus, she **(not / arrive)** on time.

5. My brother **would have bought** a sports car if he (have)the money.

Mathematical tools: Telephone numbers; Digit, Numeral and Figure

Telephone numbers

The telephone number 06539448 is said like this; oh-six-five-three-nine-four-four (double four) eight.

Digit, Numeral and Figure

English words meaning the same as the French word 'chiffre' are **digit**, numeral and figure

Digit = chiffre

Digital (adj)

Digital transmission = transmission par chiffres

Digital computer = calculateur numérique

Computer Science = l'informatique

3125 is a four-digit number.

Figure = chiffre ; the word 'figure' is used to describe a number that is not exact ie an approximate number; round figures; for example $9.9 \approx 10$ (10 is a round number)

Numeral

Roman numeral = chiffre Romain

Arabic numeral = chiffre Arabe

PRACTICE

Write the following numbers in words in English:

- a. 90,778,222
- b. 83
- c. 130,587

Session4-Reading comprehension (THE MATHEMATICIANS)

Read the text carefully and answer the questions.

QUESTIONS:

1. Who are mathematicians?
2. In the text, mathematicians can be divided into two groups. Name them.
3. Where can a mathematician work?
4. What are the major skills required from a mathematician?
5. Name four distinguished mathematicians you know.

Grammar: THE USE OF SINCE - FOR - AGO

The words **since** and **for** are often confused by many English language speakers. There is a simple rule to follow to differentiate between the usages of these words.

a-Since

Is used to talk about *time from a specific period*; It is used to indicate the beginning of an action or event. And this should be an exact time (time, day, month or year)

Rule: Sub + Present Perfect+since + Beginning of an action (or point of time)

eg.: I have been living in this house *since* 2012./ They have been waiting *since* 6.am;

NB: We also use *since* with the expression “**It is ...**” and here is the rule:

Rule: It is +Duration +since + Simple Past/ Present Perfect

eg.: 1a-We haven’t heard from them for years. b-**It is** years *since* we heard from them.

2a-I have not enjoyed myself so much for years. b-**It is** years *since* I enjoyed myself so much.

b-For

For is used to talk about **a length of time**. It is used to indicate the duration of an action. So, the duration (quantity) of time is stated (expressed).

Rule: Sub + Present perfect + for + Duration (of an action)

eg.: We haven’t seen Daddy for a long time./ I have been waiting here for 3 hours.

d-AGO

It is used **to express an action or event which now belongs to the past**. It is what explains the use of the preterit. It always comes after a time expression.

Rule: **Sub + Simple past (Preterit) + duration + ago**

eg.: -Akanava came here a week ago

-Promis went to the US two days ago

PRACTICE:

Choose the word which suits to complete the following sentences:

1-He has studied French a year (since/ for/ ago)

2-Ama has not seen her parents 2011 (since/ for /ago)

3-We met on a plane 4months (since /for /ago)

4-Alex has been a musician..... a long time (since/ for /ago)

5-Three dayshis girlfriend wrote him(since/ for/ ago)

Mathematical tool: ORDINAL NUMBERS

**Most ordinal numbers are formed by adding ‘th’ to the corresponding cardinal number
eg six = sixth.**

1st ... 12th 13th19th 20th29th 30th 90th

20th twentieth

1st First 21st twenty-first

2nd Second 22nd twenty-second

3rd Third 13TH thirteenth 23rd twenty-third 30th thirtieth

4th Fourth 14TH fourteenth 24th twenty-fourth 40th fortieth

5th Fifth 15TH fifteenth 25th twenty-fifth 50th fiftieth

6th Sixth 16TH sixteenth 26th twenty-sixth 60th sixtieth

7th Seventh 17th seventeenth 27th twenty-seventh 70th seventieth

8th Eighth 18th eighteenth 28th twenty-eighth 80th eightieth

9th Ninth 19th nineteenth 29th twenty-ninth 90th ninetieth

10th Tenth

11th Eleventh

12th Twelfth

100th = hundredth; 101st = one hundred and first.

Note how the following are written: **eighth, fifteenth, twelfth, twentieth, ninth.**

You should learn to construct sentences using ordinal numbers.

Example sentences

1. Koffi will celebrate his twenty-third birth day next week.
2. Ali lives on Tenth Sreet

Practice

Write the following as ordinal numbers by adding st, rd, nd or th where appropriate: 501, 15, 99, 1032, 93, 40

TRANSLATION: Translate the passage in italic into French

Session5-Study of scientific text: WHAT IS CHEMISTRY?

VOCABULARY:

Explain the underlined words through definition, synonym, antonym or translation.

eg. slang ==> argot (argot cacéral ou de prison); prior = anterior, preceding; enclosed ==> clos (enclosed space --->un espace clos); milestone ==> étape importante

Practice

Build a sentence with each of the following : prior, undergo, milestone

TEXT:

*Chemistry is the study of atoms, molecules, and the interactions they **undergo** called chemical reactions. Through the breaking and forming of **atomic bonds**, various **compounds** change into new compounds, either producing or consuming energy in the process. Energy may be released in the form of heat or light, as in fire or explosion. By exploiting chemistry and what we know about chemical reactions. we can produce **chemicals** with useful properties.*

The word chemistry derives from the **slang term** for the older «alchemist: a «chemist.» Modern chemistry is generally thought to have begun in the 17th century, most prominently with the experiments and writings of Robert Boyle. It was Boyle that formulated Boyle's law which holds that the temperature and volume of a gas are related.

Another important milestone was the theory of Conservation of mass, developed by Antoine Lavoisier in 1783. The law of conservation of mass holds that no mass is created or destroyed in a chemical reaction-if you burn **a log** in an **enclosed container**, the container will weigh exactly as much as it did **prior** to the reaction, only it will have changed to ash. Because of his efforts in popularizing chemistry and making it more accessible to the common person, Lavoisier is often regarded as the father of modern chemistry.

Another huge **milestone** in the development of chemistry was the formulation of the atomic theory, by John Dalton around 1800. Atomic theory holds that substances are made of small individual units called atoms which link together through atomic bonds. An aggregation of bonded atoms is called a molecule. Atoms and molecules make up almost everything that we can see and interact intimately with energy, often in the form of electromagnetic radiation. It was soon realized that electromagnetic radiation can be released when the electrons in an atom transition to a lower energy state. (...)

Grammar: ACTIVE AND PASSIVE FORMS

I/-Active Voice

- I know the Minister of education.

Form: *Subject + verb + object*

II/-Passive Voice

- The Minister of education **is known** by me.

Form: *Object + be + pp of verb + by + agent*

Notes:

- Passive voice is used when the focus is on the action. It is not important to know who or what is performing the action.

e.g.: My pen **was stolen**.

In the example above, the focus is on the fact that my pen was stolen. I do not know the person who did it.

- Sometimes a statement in passive is more polite than active voice, as the following example shows.

e.g.: A mistake **was made**.

In this case, I focus on the fact that a mistake was made, but I do not blame anybody.

A-Passive sentences with two objects/ double passive

Which object to transform into a subject depends on what you want to put the focus on.

eg.:

	Subject	Verbs	Object 1	Object 2
Active	Rita	Wrote	a letter	to me
Passive	A letter	was written	to me	by Rita
Passive	I	was written	a letter	by Rita

B-Personal and impersonal passive

eg.: 1A-They build houses. P-Houses **are built**.

2A-They say that women live longer than men. P1-**It is said** that women live longer than men. P2-Women **are said to live** longer than men

C-Have/ get (something done)

e.g.: A-The mechanic repairs my car. P-I **have** my car **repaired** by the mechanic.

D-Be going to (do)

e.g.: A-Bola is going to buy new shoes. P-New shoes **are going to be bought** by Bola

E-Use of verbs

Some passive sentences are introduced by verbs such as **say, know, agree, think, believe, consider, expect, and feel**.

eg. A-People say Koffi is intelligent. P-Koffi is said to be intelligent

The villagers think that Matata is a thief.

PRACTICE

Turn into passive or active voice

1-Victor Shoencher abolished slavery. /2-New houses are being built by Adam.

3-The lazy students were being punished by the teacher. 4-Adebayor is going to score a goal.

Mathematical tool: FRACTIONS

- 1) $\frac{1}{n}$ = one-nth, $\frac{1}{2}$ = one – half, $\frac{1}{3}$ = a/one – third, $\frac{1}{4}$ = a/one – quarter ; $\frac{1}{5}$ = a/one – fifth, $\frac{1}{8}$ = an/one – eight. $\frac{1}{100}$ = one – hundredth
- 2) $\frac{m}{n}$ = *m – nth*s; $\frac{3}{4}$ = *three – quaters*; $\frac{4}{5}$ = *four – fifths*; $\frac{7}{10}$ = *seven – tenths*; $\frac{2}{100}$ = *two – hundredths or two over hundred*.
- 3) $y\frac{m}{n}$ = *y and m – nth*s; eg $3\frac{3}{4}$ = *three and three – quaters*;
- 4) $5\frac{1}{2}h$ = *five and a half h/hours*.

Learn to construct sentences using fractional numbers:**Sample sentences**

- 1) We waited at the doctor's office for three and a half hours
- 2) The manager paid only one-third of our monthly salary.
- 3) There are 20 people in this room, 19 are students and the twentieth is a professor.

PRACTICE

Exercise 1

Write in letters in English.

- 1) $P\frac{m}{n}K$
- 2) $8\frac{10}{7}$
- 1) $\frac{1}{10,000}S$
- 3) $13\frac{5}{6}$

Exercise 2

Write in numbers

- 1) Seven and four-sixths
- 2) Nine-tenths or nine over ten
- 3) Thirteen and ten-thirteenths

Session6-Reading Comprehension (WHAT IS CHEMISTRY?)**Read the text carefully and answer the questions.****QUESTIONS:**

- 1) What is Chemistry?
- 2) What is a chemical reaction?

3)When did Modern Chemistry begin?

4)Who is Antoine Lavoisier?

5)a-Name the theories that appeared in the text. b-State their authors and define them.

Grammar: Comparatives and Superlatives

I-Comparative adjectives

We use comparative adjectives to show change or make comparisons:

This car is certainly **better**, but it's much **more expensive**.

I'm feeling **happier** now.

We need a **bigger** garden.

We use *than* when we want to compare one thing with another:

She is two years **older than** me.

New York is much **bigger than** Boston.

He is a **better** player **than** Ronaldo.

France is a **bigger** country **than** Britain.

When we want to describe how something or someone changes we can use two comparatives with *and*:

The balloon got **bigger and bigger**.

Everything is getting **more and more expensive**.

Grandfather is looking **older and older**.

We often use the with comparative adjectives to show that one thing depends on another:

*The **faster** you drive, **the more dangerous** it is.*

(= When you drive faster, it is more dangerous.)

*The **higher** they climbed, **the colder** it got.*

(= When they climbed higher, it got colder.)

II-Superlative adjectives

We use *the* with superlative adjectives:

It was **the happiest** day of my life.

Everest is **the highest** mountain in the world.

That's **the best** film I have seen this year.

I have three sisters: Jan is **the oldest** and Angela is the youngest.

PRACTICE**1-Complete the sentences with the correct form of the adjective in brackets:**

- a. Carol is as (good) as you at sport.
- b. We like wearing the (late) fashion.
- c. These trousers are _____ (comfortable) than those jeans.
- d. You are the _____ (pretty) girl in class.

Mathematical tools: DECIMAL NUMBERS

In English we use a “point” or “dot” (.) to indicate a decimal number; and after the decimal point we say each number (separately) one by one.

eg

a) 0.2 =nought point two,

b) 10.7853 = ten point seven eight five three,

c)0.0065 = nought point nought nought (or double nought) six five.

PRACTICE

Write the following in letters in English

1) 39.6

2) $0.0206 + 2.007 = 2.0276$

3) 19,220

4) $40.68 - 25.43 = 15.25$

TRANSLATION: Translate the passage in italic into French

Session7-Study of scientific text: DIFFERENT AREAS OF PHYSICS

VOCABULARY:

Explain the underlined words through definition, synonym, antonym or translation.

eg.

tumble = fall; **purview** = view, scope; **continental drift** ==>dérive continentale; a **pacemaker**==>un stimulateur cardiaque; **tiny** = very small; **decade** = a period of ten years; **unproven** = unverified; **breakthrough** = discovery

Practice

Build a sentence with each of the followings : decade, tiny, unproven

TEXT:

Physics applies scientific methodologies to understand the most fundamental principles of nature: matter and energy, and how they interact. A physicist usually specializes within one area of physics, though many disciplines overlap. Below are a few examples of the different areas of physics, with very general definitions.

Astrophysics and cosmology are examples of sciences charged with **unravelling** the mysteries of life as they play out on a massive scale. From black holes to **supernovas**, it is the job of this branch of physics to explain the birth of stars, the **tumble** of galaxies, and the beginning of our universe. Geophysics deals with understanding the earth from the standpoints of electromagnetic, seismic, and radioactive phenomena. Plate tectonics, geology, atmospheric science, climatology, and oceanography are just **a handful** of the sciences that fall under the **purview** of geophysics. Thanks to this branch of physics, we have a better understanding of events like earthquakes, volcanic eruptions, the formation of mountains, and **continental drift**.

Atomic and Nuclear Physics, along with their various disciplines, undertake one of the most fundamental areas of physics; understanding the atom. Nuclear power, radioactivity, medical imaging systems, **pacemakers**, and the atomic clock are just a few benefits derived from these branches of physics. Studying the atom also led directly to the emergence of quantum physics.

Quantum Physics is perhaps the most fascinating branch of physics, as it reveals a world so foreign to the one we know, it is all but unbelievable. The job of this area of Physics is to take us into the **tiniest** particles known to man, unlocking the most fundamental levels of matter and light. (...)

As for theoretical Physics, it rides the forefront of pioneering hypotheses for unsolved questions in all areas of physics. In some circles, theoretical Physics is not well received, because the theories are **unproven**, and there are often no tools to do so for **decades**, if ever in the scientist's lifetime. However, circumstantial support and mathematical consistency led to a **breakthrough** in superstring theory that might be the so-called *Theory of Everything*. (...)

Grammar: Comparatives and Superlatives (continued)

III-How to form comparative and superlative adjectives

We usually add **–er** and **–est** to **one-syllable words** to make comparatives and superlatives:

<i>Old</i>	<i>Older</i>	<i>the oldest</i>
<i>Long</i>	<i>Longer</i>	<i>the longest</i>

If an adjective ends in –e, we add –r or –st:

<i>Nice</i>	<i>Nicer</i>	<i>the nicest</i>
<i>Large</i>	<i>Larger</i>	<i>the largest</i>

If an adjective ends in a vowel and a consonant, we double the consonant:

<i>Big</i>	<i>Bigger</i>	<i>the biggest</i>
<i>Fat</i>	<i>Fatter</i>	<i>the fattest</i>

If an adjective ends in a consonant and –y, we change –y to –i and add –er or –est:

<i>Happy</i>	<i>Happier</i>	<i>the happiest</i>
<i>Silly</i>	<i>Sillier</i>	<i>the silliest</i>

We use *more* and *most* to make comparatives and superlatives for most two syllable adjectives and for all adjectives with three or more syllables:

<i>Careful</i>	<i>more careful</i>	<i>the most careful</i>
<i>Interesting</i>	<i>more interesting</i>	<i>the most interesting</i>

However, with these common two-syllable adjectives, you can either add –er/–r and –est/–st or use *more* and *most*:

<i>common</i>	<i>narrow</i>
<i>cruel</i>	<i>pleasant</i>
<i>gentle</i>	<i>polite</i>
<i>handsome</i>	<i>simple</i>
<i>likely</i>	<i>stupid</i>

eg.

He is certainly *handsomer than* his brother.

His brother is handsome, but he is *more handsome*.

She is one of *the politest* people I have ever met.

She is *the most polite* person I have ever met.

The adjectives *good, bad* and *far* have irregular comparatives and superlatives:

<i>Good</i>	<i>Better</i>	<i>the best</i>
<i>Bad</i>	<i>Worse</i>	<i>the worst</i>
<i>Far</i>	<i>farther/further</i>	<i>the farthest/furthest</i>

PRACTICE

2 Complete the following sentences with the appropriate comparative or superlative forms

- (a) An Ocean is a Sea (large)
- (b) Afi is Tanya (beautiful)
- (c) The Mercedes Benz is car in Lome (expensive)
- (d) Test all the students and choose (good)
- (e) I prefer this chair to the other one. Its comfortable
- (f) Everest is mountain in the world (high)
- (g) We had a great holiday. It was one of the holiday we've ever had (enjoyable)

Mathematical Tools: Percentages, Number of Times, Multiples

PERCENTAGES

25% =twenty-five percent, 30.56% = thirty point five six percent.

eg. Twenty –five percent of the students in this class are girls.

NUMBER OF TIMES

These expressions are used to say the number of times something happens.

Once, twice, thrice (or three times), n times, two and a half times

a- N times a day b-N successive times

c-N times as many or as much d-N at a time e-Increases n times or n fold

MULTIPLES

Single, Double (two fold or 2-fold) , Triple (three fold or 3-fold)

SPECIAL EXPRESSIONS**Expressions with ‘once’**

Once upon a time, Once more, Once again, At once

PRACTICE

Build sentences with the expressions: single, two at a time, once

Session8-Reading Comprehension: (DIFFERENT AREAS OF PHYSICS)

Read the text carefully and answer the questions.

QUESTIONS

- 1) Which discipline studies the origin of the world?
- 2) What benefits does humankind derive from Atomic and Nuclear Physics?
- 3) Why is Quantum Physics said to be the most charming branch of Physics?

Grammar: Infinitive and Gerund**A-We use Infinitive****After an adjective**

Eg: The new computer is really *easy to use*.

After certain verbs (with to)

Eg: 1-He *refused to pay* the bill. 2-I *want to travel* abroad.

- Here are some of the verbs → afford, refuse, agree, fail, want, help (to), plan, hesitate, prepare, would like, would prefer, would love, decide, learn, promise, choose, hope, pretend...etc

B-We use Gerund**As the subject of a clause**

Eg: 1-**Cycling** is good for your health. 2-**Drinking** sodabi is not good.

After all the prepositions except «but» and «except»

Eg: 1-I did my homework *before going out*. 2-*By hearing* the bad news, she cried.

Here Are Certain Verbs Followed by Gerund

Eg: 1- I enjoy **cooking**. 2- They suggest **going** to the cinema

- Admit, can't stand, finish, advise, deny, keep, suggest, allow, dislike, mind, waste time, avoid, enjoy, can't help, permit, stop, like, hate, detest ...

Verbs With Prepositions Followed by Gerund

Eg: 1-I'm *looking forward to* **seeing** you again soon. 2-Solo is *fond of* **listening** to music.

- Accuse of, blame for, dream about/of, look forward to, agree with, feel like, object to, apologize for, carry on, think of, ask about, complain about, give up, succeed in, believe in, concentrate on, prevent from, insist on, use for, be used to, keep on, etc

Words followed either by Infinitive or Ing-Form

Words with the same meaning

Eg: 1-I started **to read**. / I started **reading**.

- Attempt, can't bear, hate, prefer, begin, cease, intend, start, bother, continue, love.

Words with the same meaning but different use

Word	Infinitive - with an object	Gerund – without an object
advise	I advise you to go by bus.	I advise going by bus.
allow / permit	He allowed her to take the car.	He allowed taking the car.
forbid	She forbids us to smoke.	She forbids smoking.

Words with a different meaning

Word	Infinitive meaning	Gerund meaning
forget / remember	with regard to the future <i>Remember to switch off the lights.</i> Spanish: Recuerda apagar las luces	with regard to the past <i>Do you remember switching off the lights?</i> Spanish: ¿Recuerdas haber apagado las luces?
go on	start something new <i>Go on to read.</i> Spanish:	continue with the same action <i>Go on reading.</i>

regret	with regard to the future <i>I regret to say that's wrong.</i> Spanish: Lamento decir que está mal.	with regard to the past <i>I regret saying that.</i> Spanish: Lamento haber dicho que está mal.
stop	interrupt another action <i>I stopped to smoke.</i> Spanish: Paré para fumar.	terminate <i>I stopped smoking.</i> Spanish: Paré de fumar.
try	do something complicated <i>Try to solve this riddle.</i> Spanish: Intenta resolver este acertijo.	do it and see what happens <i>Try talking to him.</i> Spanish: Prueba a hablar con él.

PRACTICE**Gerund or Infinitive Exercise**

Fill the gaps with the verb in brackets in the appropriate form.

- 1- I wouldn't like _____ in his shoes. (to be)
- 2- I **hate** _____ the shopping on Saturday. (to do)
- 3- Mrs Leith offered _____ us to the airport. (to take)
- 4- My Mum demanded _____ the manager. (to see)
- 5- We like Galicia so much that we keep _____ back there. (to go)
- 6- When we visit my aunt, they expect me _____ on my best behaviour. (to be)

Mathematical tools: Mathematical Tools: SAYING THE TIME

The student is expected to say the time and date orally and to write them correctly in words in English.

What time is it? It's ten o'clock, It's ten oh five by my watch, It's nine pm, It is about nine, My clock is fast/slow, The alarm clock rings.

Do you own a clock or a watch?

eg.

- 1) Afi is taking the midnight plane to Paris (the plane the leaves at **exactly** midnight)
- 2) Ali is taking the seven-thirty bus to Kara (the bus that leaves at **exactly** seven-thirty).
- 3) The football match starts at 3 pm

It is:

7.00 Seven o'clock

7.05 Seven oh five or five minutes past seven

7.30 seven thirty or half past seven

7.45 seven forty - five or a quarter to eight

7.53 seven minutes to eight or seven fifty -three

12.00 twelve am or noon or midday

7.15 seven fifteen or fifteen past seven or a quarter past seven

13.00 one pm

Midnight, twelve pm

Notes:

am = ante meridiem = in the morning

pm = post meridiem = in the evening

use "o'clock" only on the hour (eg nine o'clock), don't use it with am or pm

Practice

Write the time it is in words: 8.15; 6.55; 1.5; 18.00

TRANSLATION: Translate the passage in italic into French

Session9-Study of scientific text: SOLAR POWER

VOCABULARY:

eg.

Solar power ==>énergie solaire; **huge** = immense, large, big; **beame down** ==>téléporter,
major ≠ minor

TEXT:

Solar power is energy generated from the sun. Many electronic devices, such as watches and calculators, can use the sun's energy directly to provide the power they need. Light energy from the sun changes the electrical conducting properties of the silicon crystals, and a tiny

electric current starts to flow. This system is called a solar cell. Although solar cells used on earth do not provide much power, satellites in space run on the same principle. They get many times more energy because they are **closer** to the sun. In the future, workers in space may build **huge** power stations from solar cells many kilometers wide. The electricity generated could be **beamed down** as microwaves and then converted back into electricity. Most of the solar power that we use today is based on a much simpler principle than the silicon solar cell. Solar panels on the roofs of houses **heat** water directly for bathing and central heating systems. The industrial version of the solar panel is the solar furnace, in which huge, curved solar panels, together with a system of mirrors, concentrate a large amount of solar energy onto a small area. The heat energy makes steam for generating electricity. Solar power is clean, renewable, non-polluting and does not damage the environment. It is potentially one of the most important sources of energy in the world. *A **major** disadvantage of solar energy is that the amount of energy generated depends on the season, the part of the world and the weather on a particular day. Another disadvantage is that the raw materials for solar panels, such as glass and aluminium, are quite expensive.*

TRANSLATION: Translate the passage in italic into French

Comprehension: Read the text carefully and answer the questions.

QUESTIONS

- 1-What is: a-Solar power? b-a solar cell? c-solar furnace?
- 2-Name four (4) advantages and two (2) disadvantages of Solar power.
- 3-Why do satellites in space get far more energy than solar cells although they run on the same principle?

Grammar: EXPRESSING CONTRAST

Contrast is expressed by terms of opposition or restriction such as: ***although, though, even though, in spite of, despite, but, yet, however, whatever, whenever, no matter, nevertheless...***

Expressions of contrast require two clauses:

eg.: 1a-Jo is wealthy, **but** he is very stingy.

b- **Although** Jo is a **wealthy** man, he is very stingy.

c- **In spite of / Despite** Jo's **wealth**, he is very stingy.

d- **In spite of / Despite the fact that** Jo is **wealthy**, he is very stingy.

Form:

Despite/ in spite of/ due to/ because of/ owing to + noun/ gerund

Despite/ in spite of/ due to/ owing to + the fact that + sub + verb (+ adj ...)

Although/ though/ even though + sub + verb

Additional information:

Whatever =====> No matter what Wherever =====> No matter where

Whenever =====> No matter when Whoever =====> No matter who

However =====> No matter how

e.g.: **Whoever** you are, you must respect the school regulations.

=> **No matter** who you are, you must respect the school regulations.

PRACTICE

Exercise1: Put one suitable word in each space

- a. Afi was short of moneyshe lent some to his sister.
- b. the heavy rainfall, the football match went ahead.
- c. Koffi went running he knew he couldn't win.

Exercise2: Complete sentence 'b' so that it has the same meaning as sentence 'a'

1a-Although it was very hot, we went to the supermarket.

b-Despite

Exercise3: Complete the sentences with: whatever, whichever, however, wherever, whoever, whenever

- a..... I see Afi, she is always in a good mood
- a. The Police will find the criminal he is hiding.

Exercise4: Fill in the spaces with one of the following words or compounds: where, however, who, which, whatever, how

- a. strong Kodjo is, his wife beats him
- b. No matter she tried, she could not control her temper

MATHEMATICAL TOOL: GIVING AND WRITING THE DATE

When we write the date we can use a cardinal number like 20 or an ordinal number like 20th.

Examples

1- July 20 2020 (July twenty 2020)

2- 20th July 2020 (the twentieth of July 2020)

3- July 20th 2020 (July the twentieth 2020)

We have said the same date in **three** different ways. In speech ordinal numbers are used

eg. The twentieth of july or july the twentieth

Three ways of asking the date:

1. What's the date today? Sunday, July the fifth
2. What's today's date? Sunday, July the fifth
3. What date is it today? Sunday, July the fifth

Example sentence

Amadou was born on August 2nd 1990

This building was inaugurated on the third of August 1975

PRACTICE

Write the following dates in letters/words in English: 1960, 1781, 1878